

Homework #3

Star Schema and Interactive Dashboard for Financial Transactions

Datasets

Three CSV files are provided:

- symbols.csv: information on listed stocks, including company, sector, industry, and country.
- account-statement-1-1-2024-12-31-2024.csv: transaction log with date, transaction type, symbol, and quantity.
- country.csv: metadata about countries, including ISO codes, region, and sub-region.

Part 1 - Data Warehouse Modeling: Star Schema

1.1 Define the Business Process and the Fact Grain

Before designing the schema, explicitly define the grain of the fact table.

Fact_Transactions must represent financial transactions with the following grain:

One row in Fact_Transactions represents one transaction from the account statement file.

1.2 Identify Fact and Dimensions

Design the star schema identifying the fact table and the required dimensions.

Type	Tables
Fact table	Fact_Transactions
Dimension tables	Dim_Time; Dim_Geography; Dim_Symbol; Dim_TransactionType

For each source attribute used in the analysis, decide whether it belongs to:

- the fact table;
- one dimension table.

Avoid unnecessary duplication of attributes across dimensions.

1.3 Define Dimension Hierarchies

Define the hierarchy levels available in each dimension.

- Dim_Time: Day -> Month -> Quarter -> Year
- Dim_Geography: Country -> Region -> Sub-region

Define appropriate hierarchies for the other dimensions when applicable.

1.4 Design the Star Schema

Create the final star schema:

- define surrogate keys for dimensions;
- map fact table foreign keys correctly;
- identify measures and descriptive attributes.

Part 2 - Data Transformation and Analysis

2.1 Load and Clean the Data

Create the dataframes required by the star schema. Keep in each dataframe only the information included in your dimensional model.

Example: If you are not using sub-region in Dim_Geography, do not include it in the related dataframe.

Perform basic data quality checks:

- check missing values;
- remove unused attributes;
- verify that every transaction symbol exists in the symbols dataset;
- verify that every company country can be mapped to the country dataset.

2.2 Analytical Questions

Provide answers to five (5) of the following questions:

1. What are the top 5 sectors by number of SELL transactions in US during 2024?
2. What are the top 5 industries by number of BUY transactions in Q4 of 2024?
3. Rank all quarters of 2024 by total number of transactions (BUY + SELL).
4. What are the top 10 countries by number of SELL transactions in 2024?
5. What are the top 5 regions by total units bought in 2024?
6. What are the top 5 sectors by total units traded (BUY + SELL) in Q3 of 2024?
7. What are the top 10 symbols by number of transactions (BUY + SELL) in 2024?
8. What are the top 5 countries by number of units traded in Financial Services companies?
9. What are the top 5 industries by units traded in Asia in 2024?
10. What are the top 5 companies with the highest average units per transaction in Q2?
11. What are the top 5 sub-regions by number of transactions in Q4 of 2024?
12. What are the top 3 sectors by total number of units sold on Mondays across 2024?
13. What are the top 5 regions by number of distinct industries traded?
14. What are the top 10 industry-region pairs by total number of transactions in 2024?

Part 3 - Streamlit Dashboard

Create an interactive dashboard using Streamlit. The dashboard must contain the following pages.

Page 1 - Time Analysis

Filters

A date range filter with default values:

- Start date: 01/01/2024
- End date: 31/12/2024

All charts in this page must react to the selected date range.

Charts

- A line chart showing total number of transactions (BUY + SELL) over time.
- A bar chart showing the top 3 traded symbols by transaction count.
- A bar chart showing the top 5 sectors by transaction count.
- A bar chart showing the top 5 industries by transaction count.

Page 2 - Country Analysis

Filters

A country selector to filter transactions by the country associated with the traded stock.

If no transactions exist for the selected country, display an appropriate message.

Charts

- A line chart showing the trend of total transactions (BUY + SELL) for the selected country during 2024.
- A bar chart showing the top industries by BUY transactions.
- A bar chart showing the top industries by SELL transactions.

Expected Deliverables

1. Jupyter Notebook containing ETL logic, star schema creation, and analytical queries.
2. A .zip file containing the Streamlit application.
3. A diagram of the star schema.
4. A brief report summarizing modeling choices, data quality issues found, and main analytical results.